



Chinese foreign acquisitions aimed for strategic asset-creation and innovation upgrading: The case of Geely and Volvo Cars

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ABSTRACT

The acquisition and integration of Western companies into Chinese business groups, the impact of host country region innovation dynamics, and the consequences for Chinese firms' innovation capabilities represent a void in the existing literature. This study applies an in-depth longitudinal case study of a cross-border M&A and links the strategic asset-augmentation literature and location theories to address the influence of target firm characteristics on the acquiring firm's innovation capabilities. Results show how the acquisition decision is influenced by post-acquisition accessibility to, and leverage of, firm-level capabilities, macro-level conditions, and the ability to draw advantages from host country regional innovation dynamics to leverage global innovation capabilities. The study extends the asset-augmentation literature to include the notion of strategic asset-creation as a motive for M&A and demonstrates the dynamics of strategic asset-creation in the automotive industry. This study makes three contributions. First, it provides a definition of strategic asset-creation in the automotive industry; Second, it provides a more comprehensive explanation of the importance of spatial dynamics in regional attractiveness in strategic asset-creation processes of emerging market MNCs (EMNCs); Third, it demonstrates how EMNCs harness, absorb, and create new strategic assets.

1. Introduction

Chinese industries' continued integration into the world economy is a relatively new contributor to regional economic development that has emerged during the last decade (e.g., Alon et al., 2012; Alvstam et al., 2009). Being among the most important trading partners for most developed economies, China has succeeded in attracting foreign direct investment (FDI) from many developed economies. This inward FDI has enabled China to attain technological capabilities primarily in manufacturing, and Chinese firms have increasingly become more skilled in terms of utilizing advanced manufacturing technology in industries such as the automotive industry (Alvstam et al., 2014).

The development path of China via inward FDI follows the mainstream theories on FDI (e.g., Blomström and Kokko, 2003; Borensztein et al., 1998; Enderwick, 2005). For many years, attracting foreign investment has been promoted as the standard recipe for economic growth and development (e.g., Ruane and Ugur, 2006; UNCTAD, 2001, 2002, 2013), which is particularly true for China. Western FDI in China is, unsurprisingly, well documented in the literature (e.g., Delios et al., 2009; Isobe et al., 2000; Zhou et al., 2002). However, as economic development has progressed in China, the realization that innovation capabilities are crucial in attaining higher echelons of competitiveness

has grown among Chinese manufacturers (Liu and Woywode, 2012; Mathews, 2006, 2009). For Chinese manufacturers in knowledge-intensive industries producing products with high-technology content, foreign company acquisitions have become important vehicles for rapidly obtaining innovation capabilities. Considering developments in the Chinese economy and the increasingly important role Chinese outward investments plays in Western economies, there is an obvious need for more research on the global and regional integration of Chinese firms' innovation capability build-up (Alvstam et al., 2014). This paper investigates the interface between existing firm innovation capability, cross-border acquisitions as a rapid means of creating new core innovative capabilities, and the geography of Chinese automotive firms' internationalization. Specifically, this paper draws upon the experience from the Zhejiang Geely Holding Group (Geely) acquisition of Volvo Cars Corporation (Volvo) to explore how Geely is using Volvo's capabilities to create new automotive innovation capabilities that integrate the two companies not only at the regional level but also at the global level.

Locational issues are an important topic in International Business (IB) studies (e.g., Beugelsdijk et al., 2010; Schlunze et al., 2011; Beugelsdijk and Mudambi, 2013), but attempts to explain location decisions of FDI (e.g., Dunning, 1992; Rugman, 1981) have taken a rather

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static perspective and been limited to competitive advantages of host nations. Concurrently, the influence of geographic distance and factors in global trade patterns (e.g., Santos Silva and Tenreyro, 2003) on economic geography and international trade has been discussed since the 1990s (e.g., Fujita et al., 1999; UNCTAD, 2001). Furthermore, at the intersection of economic geography and international business, regional locational dimensions have been suggested as important factors (e.g., Beugelsdijk et al., 2010; Ström and Wahlqvist, 2010). More recently, the increased importance of innovation networks and complex relations that are often found at the regional level make analyses at the national level increasingly irrelevant (e.g., Beugelsdijk and Mudambi, 2013; Johanson and Vahlne, 2009).

Furthermore, within economic geography and innovation studies, the role of national and regional innovation systems has held a central position in theorizing and explaining the competitive advantage of nations, regions, and ultimately firms (Asheim and Gertler, 2006; Asheim et al., 2011; Dicken, 2015; Lundvall, 1992). Amighini and Franco (2013) observe an agglomeration pattern among the emerging market multinational corporations' (EMNCs) technology-driven innovation investments, which tend to be drawn to regions with a concentration of knowledge-intensive high-technology firms. These agglomeration effects have also been important for the regional development of company or knowledge clusters in both manufacturing and services industries (Ström and Wahlqvist, 2010). Thus, globalization has led to increased awareness of the locational factors among firms in establishment decisions, where global production chains are established along the most favorable economic locations based on either cost or knowledge. By offering special expertise and labor skills, favorable locational advantages tend to attract knowledge-intensive investments regardless of the physical distance between host and home countries.

Processes that still need to be studied include FDI location decisions, utilization of target firm capabilities, and the influence of geography and host region innovation dynamics. These processes are especially interesting when Chinese outward FDI is taken into consideration (Chaminade and Rabelotti, 2015). For instance, whereas innovation as a core dynamic capability among knowledge-intensive high-technology firms has been discussed in the literature for many years, it has often been treated from the narrow paradigm of resource-based view perspectives (e.g., Barney, 1991) mostly from Western multinational corporations' perspectives, which risk missing out important properties of how strategic innovation resources are supposed to be attained.

In addition, quantitative analyses dominate the research field on mergers and acquisitions' (M&As) role in developing research and development (R&D) and innovation capabilities (e.g., Bertrand and Zuniga, 2006; Blomström and Sjöholm, 1999; Buckley et al., 2002), making it difficult to reach inside the "black box" and to know how firms succeed in creating (dynamic) innovation capabilities. Hence, qualitative studies on how cross-border M&As benefit investing foreign firms, and especially Chinese multinational corporations (MNCs), are less frequent and represent an important research gap (e.g., Deng, 2009; Ström and Nakamura, 2014).

1.1. Purpose and research question of the study

The purpose of this study is to analyze qualitatively strategic asset-creation by an emerging market automotive firm. More specifically, we address how such asset-creation contribute to increased innovation capabilities, and how strategic asset-creation is augmented by the Chinese firms FDI location and utilization of a Western target firm's technological portfolio. Asset-creation and innovation capabilities are studied through an in-depth longitudinal case study, which shows the inherent complexity of asset-creation initiatives while highlighting the organizational imperatives used by the involved firms. In this study, the discussion is limited to the experiences from the Zhejiang Geely Holding Group (Geely) acquisition of Volvo Cars Corporation (Volvo).

Our choice of case is highly relevant not only because it represents a Chinese strategic asset-seeking overseas investment, but also because this case involves a high-profile acquisition of an established Western automotive maker with a strong global brand and high level of R&D investments. What makes this specific case interesting is its uniqueness in presenting the creation of a new, jointly shared, strategic asset that leverages existing strengths from both Volvo and Geely Auto. Therefore, this paper aims to answer the question "*How are innovation capability leverages created in Chinese automotive firms from foreign acquisitions, and how are the spatial dimensions utilized by Chinese investors in this process?*"

In this paper, FDI is defined as cross-border M&A, not only because FDI is a recognized mode for development (e.g., Blomström and Kokko, 2003; Borensztein et al., 1998; Enderwick, 2005; UNCTAD, 2013) and cross-border M&As are appropriate for detailed analyses of FDI cases (e.g., Ström and Nakamura, 2014), but also because the majority of firm-level Chinese outward FDI (OFDI) are cross-border acquisitions (e.g., Ren et al., 2012). Innovation capability is defined along the technological trajectory, where innovation capabilities refer to a firm's ability to enhance its technological innovativeness and create new business value propositions by acquiring, utilizing, and developing valuable technological resources and knowledge (Burgelman et al., 2004).

This paper is organized as follows. Next, the theoretical framework and methodology will be discussed, followed by the case description and analysis of the joint independent R&D company of Geely Auto and Volvo called China Euro Vehicle Technology (CEVT). After discussing the results and their implications for practice and theory, we wrap up with a brief concluding discussion.

2. Theoretical framework

The internationalization of Chinese firms has received due attention from scholars who have attempted to explain the underlying rationale for this expansion and the strategies used by different kinds of firms (e.g., Amighini et al., 2010; Child and Rodrigues, 2005; Morck et al., 2008; Wang et al., 2012). The Chinese manufacturing sector represents a concrete example of China's strength in production and industrial ownership integration, gained by expanding its regional production networks to include Western firms and production units. In earlier studies (e.g., Alon et al., 2012; Ström and Nakamura, 2014), Chinese firms that acquire foreign companies are characterized as being at an early internationalization stage. Typically, these M&As have occurred only after implementation of the Chinese "going out" policy in 2000 (Ström and Nakamura, 2014).

A mainstream approach to theorize about Chinese foreign investments arises mainly from resource-based perspectives (e.g., Barney, 1991). Dunning's (1992) and Rugman's (2010) theoretical frameworks are examples of attempts to explain FDI motives from resource ownership, resource access, and locational aspects. Deng (2004), Hong and Sun (2006), and Buckley et al. (2007) follow this line of argument by suggesting diversification and strategic asset-seeking, in addition to mainstream motives such as resource and market-seeking investments, as the main overseas investment motives of Chinese firms. Ström and Nakamura (2014) add to this discourse by arguing that the motive for foreign investments is not primarily to establish a presence in a foreign market but to strengthen Chinese firms' competitive position in their own growing home market, which is consistent with Mathews' (2006) findings. Other studies on Chinese OFDI to industrialized countries and regions have shown that Chinese investors are seeking strategic assets, among other aims, to enhance their innovative capabilities (e.g., Alon et al., 2012; Alvstam and Ivarsson, 2014). Acquired host country firms can, therefore, provide positive effects in terms of innovation capabilities for foreign investing firms (Fu and Zhang, 2011; UNCTAD, 2006; Wang et al., 2012).

2.1. Spatial dynamics of regional attractiveness and cross-border M&A as a vehicle for innovation development

Trade and FDI is the result of how firms see opportunities provided by specific locations when formulating their corporate strategies, which determine the preferred M&A mode (Dunning, 1992). This cross-border M&A activity can be interpreted as resulting from an ambition to create networks and relations between the home and host countries (Bathelt and Glückler, 2003) where related industries can build regional economic development over time (Neffke and Henning, 2013).

Anand et al. (2005) point to capability improvements in R&D, quality, and product line extension as a result of cross-border M&As due to the acquirer's resource redeployment over the geographic distances of the newly acquired and diverse resources. Thus, networks that span mature and emerging markets are a result of the will to optimize resources over space and utilize different locational advantages, such as a low-cost base in emerging market home countries and specific knowledge pools in advanced host economies. Regional industrial structures can facilitate the assets sought by the investing firms, such as knowledge or competence within an industry, or the value of company brand names already established in the global market (Alvstam et al., 2014).

Networks and relationships play an important role in the establishment of firm-level competitiveness and innovation (Bathelt and Glückler, 2003). Firms seek different types of location-specific advantages that can be utilized for the overall firm strategy. To build an international presence, location can be used for market access, asset-seeking, and global brand awareness. Multinational firms are important in the diffusion of innovation in the global market, where they are at the core of trade, investment, and collaboration among network actors (Narula and Zanfei, 2006). A region with strong innovation systems can facilitate this kind of positive economic development and also be in a better position to manage external shocks from changing industrial structures in the global market (Sölvell et al., 2015). This can involve a combination of specific location-bound knowledge bases, but also how a related variety of industries creates regional advantages (Asheim et al., 2011). These regional advantages can then be transformed into firm-level competitive advantages. Firms can use the host market for different purposes. FDI in the form of M&A can be seen as a vehicle for leapfrogging or a springboard to the technological development ladder through buying more advanced firms in mature economies (Alvstam and Ivarsson, 2014; Luo and Tung, 2007). The establishment of new firms is yet another way to take advantage of these locational advantages. These establishments can develop a knowledge-absorption character by combining home market and host market capabilities. EMNCs have successfully used various strategies, related to what Mathews (2006) attributes to an ability to create leverages, create linkages, and absorb knowledge, within these internationalization processes. Firms from these markets can employ host market advantages in combination with home market advantages to build competitive advantage. This further relates to how firms use institutional environments for their strategic development.

2.2. Strategic asset-seeking and asset-augmentation

EMNCs' strategic asset-seeking FDI in developed economies has been mainly driven by competitive disadvantages with regard to the acquiring firms' innovation capabilities (Cui and Jiang, 2009). Given that Chinese firms' advantages are mainly based on country-specific advantages (CFAs) rather than firm-specific advantages (FSAs), strategic asset-seeking motives of Chinese firms investing abroad involve both obtaining foreign technologies and enhancing existing technological capabilities (Rugman and Li, 2007; Wang et al., 2013; Gugler and Vanoli, 2015).

More recently, in their strategic asset-seeking, EMNCs' motives increasingly aim to augment home country strategic assets with foreign assets, a perspective that has gained increased traction within the

EMNC literature (e.g., Yadong and Huaichuan, 2009; Wei, 2010; Deng, 2012). Referred to as the "asset-augmentation perspective" (Buckley et al., 2016, p. 2), the argument is that since EMNCs lack the competitive advantage required to out-compete local firms in foreign markets, they instead seek to build competitive advantage by augmenting specific strategic assets and resources.

Strategic asset-seeking motives involve augmenting areas of perceived competitive disadvantage through the acquisition of a variety of intangible assets, such as technologies, R&D capability, managerial competency, human capital, and brand names (Dunning and Lundan, 2009; Ferreras-Méndez et al., 2016; Sun et al., 2012). Thus, with regard to strategic asset-seeking motives, such as those concerned with obtaining innovation, R&D, or technological capabilities, asset-augmentation is central (Dunning and Narula, 1995; Alvandi et al., 2014). This also implies that EMNCs at times acquire foreign firms with the sole purpose of accessing and augmenting, rather than exploiting, ownership advantages (Dunning, 1998).

Therefore, asset-augmentation through M&A represents activities whose prime purpose is to create new knowledge that augments existing competencies, with the ultimate aim of building competitive advantages within the acquiring firm (Buckley et al., 2016). This can be done through the acquiring firm's own (formal) R&D activities or through other non-hierarchical means in partnership with other economic actors (Narula, 2010). In other words, by acquiring and internalizing existing assets (Voudouris et al., 2012) or by creating completely new technological assets through foreign-located facilities (Narula, 2006), the acquiring firm aims to enhance its own innovation capabilities (Cuervo-Cazurra, 2012).

A number of authors have claimed that asset-augmentation motives conform more closely to the perspective of latecomer motives for engaging in an acquisition, where exploitation of existing ownership advantages is secondary to technology and innovation capability catch-up (e.g., Child and Rodrigues, 2005; Yiu et al., 2007; Wei, 2010). This stands in direct contrast to more traditional views that the effects of M&As on Chinese investor firms are better understood as the benefits of exploiting existing knowledge held by Western partners without any technological upgrading (Deng, 2009). The underlying rationale driving the asset-augmentation strategy is that the foreign location provides access to location-specific advantages that are not as easily available in the home base (Narula, 2006). Subsequently, when the strategic motive is asset-augmentation, the R&D operations or activities of acquiring EMNCs are often located in knowledge-intensive countries (Mathews, 2006; Yiu et al., 2007). The literature on location advantages, for instance, identifies that augmenting technological competence is a matter of not only the MNC's current resource possession but also of the efficiency by which it utilizes complementary resources associated with the relevant local innovation system (Narula, 2006). This has led to claims that asset-augmenting activities require proximity to the economic units and innovation systems from which they seek to learn (Criscuolo et al., 2005; Narula, 2006). However, few studies have examined the "black box" inside these firms, that is, how the results of these augmentation strategies in terms of innovation capabilities are acquired through strategic asset-seeking investments abroad (Park and Choi, 2014; Gugler and Vanoli, 2015). It is not clear how EMNCs harness, absorb, and augment such assets (Narula, 2010), calling into question whether such firms have sufficient capabilities to properly augment their strategic asset-seeking-related acquisition (Anderson et al., 2015; Rugman and Li, 2007). This inability is explained in the literature by Chinese firms' lack of experience in R&D activities, in foreign M&As, and in internal managerial knowledge, as well as their lack of high-skilled labor and high-tech knowledge (Rugman and Li, 2007; Shenkar, 2009; Gugler and Vanoli, 2015). Given the Chinese firms' supposed lack of absorptive capacity, foreign strategic asset acquisitions do not automatically imply the active augmentation of existing ownership advantages (De Beule and Duanmu, 2012; Voudouris et al., 2012).

As discussed above, the dynamics and motives for strategic asset-augmentation differ from traditional asset-exploitation motives predominantly concerned with the optimal mode of co-ordinating and developing existing resources and capabilities (Dunning and Narula, 1995; Dunning, 2000; Alvandi et al., 2014). From an organizational learning and development perspective, for instance, exploitation and augmentation motives also differ (Li et al., 2016). More precisely, “when exploiting existing resources, acquirers are teachers; when seeking strategic assets they are students” (Li et al., 2016, p. 474). Lane and Lubatkin (1998) use the student-teacher perspective in discussing the importance of relative absorptive capacity not only for how to structure and manage learning and development but also for its importance in deciding from whom to learn. A firm's ability to learn from another firm is jointly determined by the relative characteristics of the “student” firm and the “teacher” firm. Lane and Lubatkin (1998) report that the higher relevance the acquiring firm's knowledge holds in relation to the acquired firm, the higher the inter-organizational learning or, in other words, the higher the proportion of organizational problems that the acquiring firm shares with the acquired firm, the more inter-organizational learning will take place. Relatedness has been conceptualized as the closeness or similarity between the acquired and acquiring firms' products, customers, and resources (Chatterjee, 1986; Lubatkin, 1987), although more recent studies have argued that its conceptualization should be extended to incorporate other dimensions, including cognitive, social, organizational, technological, institutional, and geographic dimensions (Boschma and Frenken, 2011).

The use of external partners to acquire new organizational capabilities for the purpose of internalizing and enhancing existing innovative abilities (West and Bogers, 2014) is often discussed in the literature in the context of firm-to-firm contractual relationships (e.g., outsourcing). Firms from emergent markets have also seized the acquisition opportunities that the “fire sales” of ailing Western firms present (UNCTAD, 2000; Ström and Nakamura, 2014). Thus, cross-border M&A has become an efficient way for EMNCs to acquire and utilize exploitation and exploration efforts of other, foreign, firms and their R&D investments. However, using cross-border M&A as a tool to create innovation capabilities is not discussed at length in the literature, and we lack a theoretical discourse that addresses the particular aspect of how international acquisitions are used to create new capabilities in connection to innovation. Thus, our understanding of the role Chinese OFDI plays in innovation enhancement and the use of cross-border M&A among Chinese MNCs to create innovation capabilities needs to be developed. With little prior qualitative and in-depth empirical research to draw on as a base, the question of how EMNCs potentially create strategic assets with multiple locational configurations is yet to be fully explored.

3. Method and empirical data collection

3.1. Longitudinal single case study

This paper pertains to research advantages obtained through a longitudinal, real-time, and deep single case study that increases

internal validity by facilitating the tracking of cause and effect (Blazewski, 2011; Pettigrew, 1995). The research strategy adopted in this study is best understood as a contextualized explanation (Welch et al., 2011) of the spatial dynamics of regional attractiveness and M&A as vehicles of innovation development (Birkinshaw et al., 2011). The aim is therefore to understand both organizational factors and the role of external factors in innovation capability development. A central assertion is that this case allows us to generate causal explanations that preserve contextual richness (Welch et al., 2011). The merits of case-based studies when attempting to explain the relevance of cause-and-effect relationships is well established (Pettigrew, 1990). This case is selected for its purposefully revelatory nature as it allows us to observe and analyze regional dynamics and the development of innovation capabilities through M&A, which to the best of our knowledge, have not been studied frequently using a qualitative approach (e.g., Deng, 2009; Ström and Nakamura, 2014).

The selected case organization is China-Euro Vehicle Technology (CEVT), which is the Chinese-owned Geely Zhejiang Holding Groups' (Geely's) most recent R&D center. CEVT's main location is in the western Sweden automotive cluster of Gothenburg. CEVT is a research center engaged in almost all facets of the vehicle development value chain, including pre-conceptualization and design, product strategy and planning, technology development, detailed engineering, prototype development and physical testing, purchasing/sourcing, and program financing. Its main responsibility and mandate lie in developing and maintaining a new vehicle platform, the Compact Modular Architecture (CMA), to be used jointly by Volvo Car Corporation and Geely Auto. Developing a new vehicle platform constitutes one of the most complex facets of the vehicle development process in terms of activities, co-ordination, responsibilities, and demands for innovation capabilities (Yakob, 2015). Thus, the case is a good example of a recent effort by a Chinese MNC (i.e., Geely) to utilize the spatial dimension of its foreign acquisition (i.e., Volvo) to enhance the group's overall innovation capabilities in general, and the automotive vehicle development capabilities of Geely Auto in particular. Concurrently, the selected case organization (i.e., CEVT) combines, integrates, synthesizes, and embodies much of the innovative capabilities needed for Geely Auto's global ambitions and the strengthening of its future global competitiveness. Fig. 1 summarizes the timeline of our study.

3.2. Data Collection

Collection of the empirical data is based on tracing key events in close interaction with the three organizations involved: CEVT, Volvo, and Geely. The empirical data were collected within a larger research project on Geely and Volvo carried out during 2011–2016. This project has followed Geely's acquisition process of Volvo and its subsequent developments. Large numbers of interviews and field study visits have been conducted throughout the six-year study. As depicted in Table 1, interviews were carried out with managers at CEVT in Gothenburg, Sweden, Volvo Car Corporate Headquarters in Gothenburg, Sweden, Volvo Car China in Shanghai, China, Volvo's vehicle assembly plant in Chengdu, China, and Geely headquarters in Hangzhou, China.

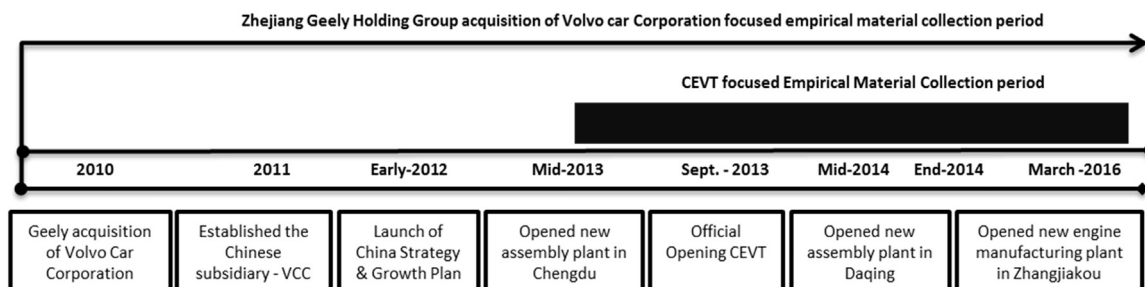


Fig. 1. The timeline of the case under study.

Table 1
Method description – Overview of empirical data collected.

Technique	No. of Interviews	No. of respondents	Interview location	Departments/functions	Topical focus (based on interviews and secondary sources)
China-Euro Vehicle Technology (CEVT)			Gothenburg	Business Office, Exterior & Body, Vehicle Integration,	Organization and Organizing, Roles and Responsibilities, Technology Competence, Technology Strategy, Knowledge & Learning Transfer, Innovation Capabilities, Collaboration, Location, Activity Scope, CMA
Deputy CEO,	2	1			
Director,	1	1			
Senior Manager	3	3			
Volvo Car Corporation			Shanghai,	HR, Powertrain, R&D Product Line	Strategy, Managerial Capabilities, Human Resources (HR), Product Portfolio, Co-ordination, Collaboration, Localizing, Scale/Scope, Location, Technological Capabilities
Vice President,	15	8	Chengdu	Management & Program Office, Purchasing, Vehicle Engineering, Marketing Vehicle Line Mgmt., Industrial Engineering	Strategy/Strategic Intentions, M&A Motivation, Collaboration & Co-ordination, Technology Strategy
Director	22	16			
Geely			Hangzhou	Corporate communications	
Director	2	2			
Total	45	31			
Documents					M&A Geely/Volvo, Establishing CEVT, Technology Development, Growth/Recruitment, Collaboration, CMA, Organization & Organizing
Corporate Presentations, Internet Newspapers					

In line with Leonard-Barton (1990), empirical data collection attempted to “slice” through multiple organizational levels and perspectives to understand the complex empirical setting under investigation. Subsequently, the respondents were drawn from several hierarchical levels at the different organizations to mitigate respondent bias (Eisenhardt and Graebner, 2007). For example, discussions about joint sourcing initiatives to accommodate shared platform components came up in interviews taking place at both Volvo headquarters (HQ) and when visiting factories in China in early 2013 and 2014, and discussions around future C-segment vehicles were covered in interviews with the Product Planning and Vehicle Line Management function of Volvo Car China in early 2013 and 2014. Furthermore, interviews conducted in late 2014 at Volvo Car China revealed that a new factory would be built in Taizhou, Zhejiang province, to accommodate vehicles based on the new CMA platform. Finally, Geely's ambitions for the new R&D center in Sweden, Gothenburg, were mentioned in interviews at Geely HQ in Hangzhou in late 2014. The case organization presented here is the first major result of Geely's ambitions to consolidate its automotive R&D capabilities on a global basis following its acquisition of Volvo in 2010.

Given the strategic outlook of our study, interview efforts were focused on the senior manager levels and above. In total, 45 tape-recorded interviews were conducted with 31 respondents, amounting to approximately 50 h of interview time. Interviews were conducted with 1 Deputy Chief Executive Officer (CEO), 8 Business Function Vice Presidents, 19 Business Function Directors, and 3 Senior Managers. The total number of interviews is higher than the number of respondents because some managers were interviewed two or three times. Interviews were carried out in both English and Swedish. Each interview lasted from 30 min to 2.5 h and followed roughly the same structure. Respondents were asked about their previous professional experience, their motives for working at their current organization, the R&D and innovation challenges faced, benefits and/or potential challenges of being located in Gothenburg or somewhere else, how regional innovation dynamics are experienced and what benefits they accumulate, and how the organizational entity has developed its R&D and innovation capabilities. Notes were taken during interviews to capture topics in the form of keywords and significant situations and events that seemed significant to the respondents. These keywords, situations, and events were further probed in subsequent interviews. All interviews were digitally recorded and transcribed verbatim, translated into English, and compiled chronologically to create a rich case illustration. Empirical data from secondary sources were used, such as internal corporate presentations, news articles, and other open source information from the Internet.

3.3. Empirical data analysis

The empirical data were analyzed thematically to isolate key findings pertaining to the subsequent development of CEVT. Coding and analysis of the collected empirical data occurred in three separate but interrelated steps. First, a constant comparison technique (Miles and Huberman, 1994) was used for inductive open coding of the empirical data, using the words of the interview participants. During the coding process, the authors used selected informant quotations to strengthen and clarify each identified concept. This exercise was first carried out independently by each author, followed by a joint discussion and comparison. This open coding was provisional and refutable unless further strengthened by means of iterating with the case data throughout the analysis process. Internal validity was secured by all three authors participating and cross-checking the inductive analysis process when the identified codes were grouped into first-order categories through iterative comparison of the various extracts. The first-order categories were distinguished and grouped together through identification of sections in the empirical data that had coherent meaning and represented a generic but important issue in the case data. Linkages between first-order categories were then sought by means of axial coding (Corbin and Strauss, 1990), resulting in a number of second-order themes that represent a higher level of abstraction based on inductive reasoning. Finally, by means of cross- and within comparison of both first-order categories and second-order themes, a limited number of aggregated dimensions were created. These aggregated dimensions were used as the basis for the theoretical implications discussed in this study. At the core of the analysis process was the step-by-step creation of a data structure that lay at the foundation of the emergent findings (see Fig. 2). The overarching aim of the coding and analysis process was to discern important aggregated dimensions that could further explain the evolution of the case organization CEVT. Hence, the desired case output was a narrative that provides closeness to first-order participant perspectives and second-order themes that cumulate in the form of a desired explanatory case.

In the analysis process, emergent evidence indicated that the interdependence among access to highly skilled labor, the institutional and cultural setting of Sweden, and the location in a competitive regional knowledge center in western Sweden gave rise to important locational/spatial dynamics and had an important influence on the location establishment decision. Subsequently, *macro-level conditions* emerged as a central aggregated dimension for innovation capability development and the creation of the strategic asset. Furthermore, existing *firm-level capabilities* were crucial in the creation of further innovation capabilities. In particular, the technological know-how of

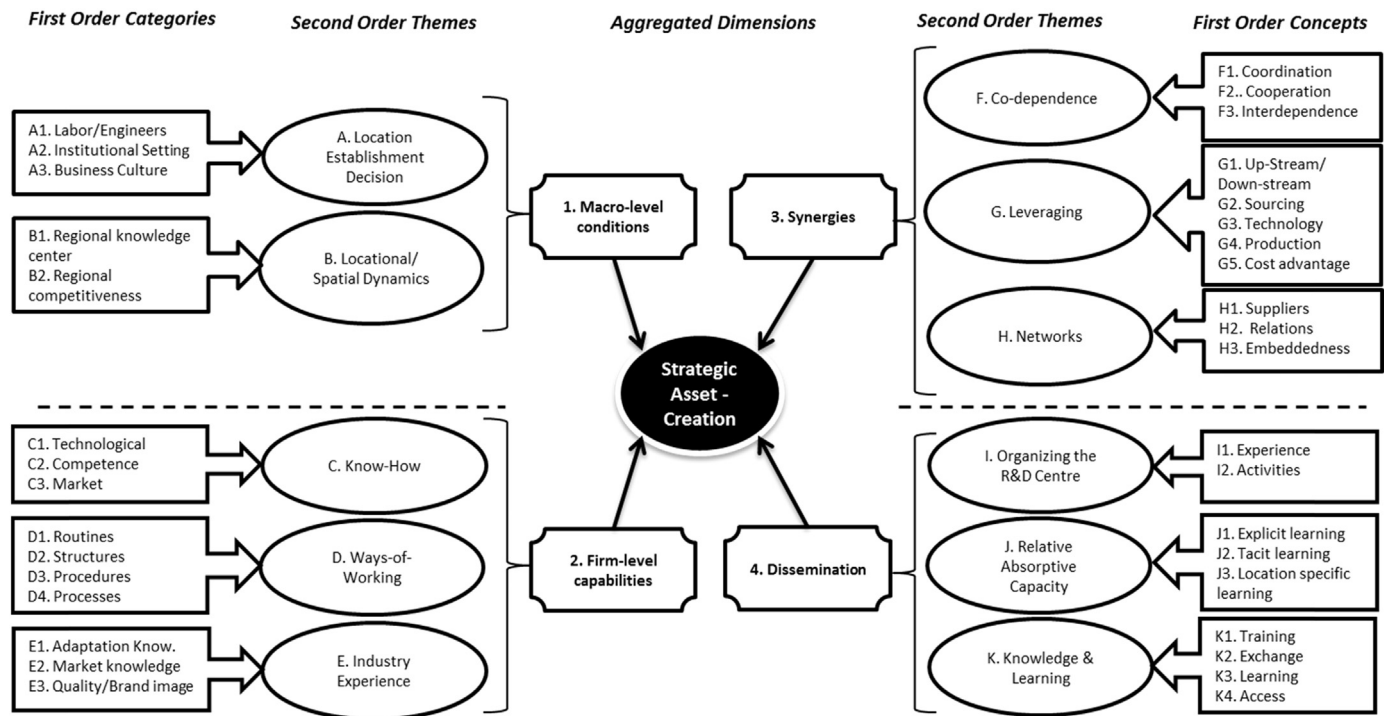


Fig. 2. Data structure of first-order categories, second-order themes, and aggregated dimensions.

Volvo, the Chinese market know-how of Geely, and highly developed ways-of-working drawing on established routines and processes could be observed as central components of this process. Furthermore, the ability to leverage sourcing demands, utilizing local cost advantages (e.g., consultants in Sweden and components in China), and the access to production facilities also stood out as important aspects of innovation capability development and the creation of the new strategic asset. It also became evident that although benefits from location dynamics were sought from the western region of Sweden, attempts to identify strengths that could be drawn from locations in China were also important. However, such benefits were of a different character, aimed toward achieving *synergetic advantages*. The means to achieve this included leveraging resources through organizational adaptation and integration between Volvo and Geely and between the CEVT entities in Sweden and China. It also included assigning lead responsibilities to either China or Sweden across the vehicle value chain and drawing on the two firms' networks with access to suppliers and established relations with stakeholders. A final mean included facilitating the handling of co-dependence through strong co-ordination and co-operation. Finally, the relative absorptive capacity of Geely in relation to Volvo, the organizational setup of CEVT, and the explicit aim to engage in knowledge transfer, development, and learning point to the importance of *dissemination*. Such dissemination was based on both tacit and explicit learning through (Chinese) staff mobility, training programs, access to more experienced Western engineers, and the gradual increase in vehicle value chain engagement of the Chinese entities and employees. Thus, the drawing on specific firm-level capabilities, macro-level conditions, synergy effects, and the dissemination of competence, knowledge, and learning, emerge as important aggregated dimensions that explain the development of CEVT.

3.4. Validity and reliability of the research and empirical data

Internal reliability (i.e., trustworthiness) of the collected rich empirical data is supported by the involvement of multiple researchers. Furthermore, respondent validation on emerging interpretations was gained throughout the longitudinal case research study. The empirical

data have been presented and discussed throughout the research process at seminars held at the authors' research unit, where researchers who have been involved in research on the same case organization have provided feedback. These seminars provided not only a forum for discussing emergent themes and aggregated dimensions but also a peer reference group for validation of the findings and interpretation of the empirical data.

4. Case: China-Euro Vehicle Technology

While many Chinese automakers have had access to Western automobile technology through joint venture constellations in the home country, few have an ownership link similar to the one between Geely Auto and Volvo. The key infrastructure to facilitate co-operation between Geely Auto and Volvo is CEVT, an independent research and development center. Work on setting up the center began in February 2013 and CEVT opened officially in September 2013. CEVT is a wholly owned subsidiary of Geely and represents an important strategic step in Geely's ownership of Volvo. As one respondent said, "CEVT constitutes the foundation for Geely's future European and global operations in the C-segment of the automotive industry." (Deputy CEO, CEVT, 2015). This foundation is based on the development of the new state-of-the-art modular vehicle platform called Compact Modular Architecture (CMA), which enables the development, manufacturing, and assembly of world-class vehicles for both Geely Auto and Volvo. This ambition correspondingly involves the creation of a global performance organization engaged in technology innovation and development, collaboration, and talent recruitment and management. The funding to establish CEVT and support R&D development predominantly comes from Geely, which has raised parts of this funding from local governments in China. The groundwork for gaining this external financial support has been carried out by Geely HQ. This also means that Geely China owns the intellectual property rights of the CMA (see Table 2 for further case details).

Table 2
Case details - China-Euro Vehicle Technology (CEVT).

Characteristics	
Contextual attributes	
Ownership	Legally foreign owned and private. High autonomy, strong mandate within the Geely Holding Group.
Location	Gothenburg automotive cluster. Proximity to/ availability of needed competence and experience. Close to Volvo. Close to technology consultancy firms.
Scope	Development and maintenance of a cross-brand, inter-organizational automotive platform for C-segment cars. Substitution economies/scale economies derived from procurement synergies and reuse of system solutions and components. Scope economies from carry-forward shared solutions.
Strategic aims	Strengthening long-term competitiveness of Geely Auto and Volvo. Develop world-class vehicle architectures/new technologies. Raise product quality/optimize resource utilization. Implement a shared platform strategy. Product quality, brand image, and cost synergies. Bring technology strengths of Volvo and the cost advantages of Geely Auto together. Important learning areas including technology, development processes, work methodologies, and how to do business in the global automotive industry.
Tactical imperative	Quickly scaling-up human capital pool. Ramp-up of technological responsibilities, intra-group knowledge sharing, division of labor, responsibilities, and activities.
Organizational Attributes	
Entities & responsibility	CEVT Gothenburg. CEVT Hangzhou. Geely Design Gothenburg. Complete single-brand responsibility (Geely Auto), integrative co-operation with Volvo for top hats. Planning, selection, development, and maintenance of platform elements. Cater to brand differences in vehicle characteristics, attributes, and cost structure. Maintaining brand integrity.
Structure	Eight departments (R&D, CMA Product Strategy & Vehicle Line Management, Quality, Finance, HR, Purchasing, Geely Design (Gothenburg), and Business Office). Eight R&D functions (Architecture, Powertrain Integration, Chassis & Safety Electronics, Interior, Exterior & Body, Vehicle Integration, Transmission, and Manufacturing Engineering).
Division of labor	Work split Geely Design, CEVT Gothenburg (Concept Phase), CEVT China (Launch Phase).

4.1. Establishment decision and the advantage of spatial dynamics

CEVT is located in the heart of Sweden's automotive cluster in western Sweden, close to key industrial actors such as Volvo, Volvo Technology, Volvo Bus, Volvo Information Technology, Ericsson, Semcon, Chalmers University, and the University of Gothenburg. The rationale for establishing CEVT in Gothenburg is built on a number of aspects. One important aspect is the close proximity and availability of needed competence. Geely, in conjunction with Volvo, early on identified the location-specific advantages stemming from the historical presence of Volvo in Gothenburg and the industry-related innovation competencies that exist in the vicinity of the company. Talking about this issue, a respondent said: "The strong competence associated with, and the regional competitiveness of, the automotive cluster in western Sweden played an important role in the establishment decision." (Deputy CEO, CEVT, 2015). Another reason is that the Swedish business culture allows for a flatter organizational structure, with a smaller managerial hierarchy, which encourages co-operation and increases the speed of decision making. Hence, the Swedish approach to organizing and management was deemed something to emulate, given the ambitious growth intentions of CEVT. Finally, following the convention in the automotive industry of using a large number of external technology consultants, Swedish consultants are deemed cheaper to use than those in many other countries when looking at the experience/hourly consultant cost ratio (Interview, Director, Exterior & Body, CEVT, 2015).

CEVT holds the same status as Volvo and Geely Auto within Geely, according to CEVT's CEO Mats Fägerhag (Just Auto, 2014) since "much of the Group's future is decided here [at CEVT]." (Ny Teknik, 2015, authors' translation). Accordingly, it was important to emphasize in the set-up process that CEVT would not be a stand-alone organization or merely a function within the Geely Group. Rather, it was important to emphasize that CEVT would become, as one respondent put it: "...an integral part of the Geely Holding Group and we make an effort to make ourselves integrated." (Director, Exterior & Body, CEVT, 2015).

CEVT was established in February 2013 with seven employees, an initial budget of RMB 4 million, and an ambition to establish an R&D organization with a target of roughly 100 employees. By the end of 2013, the center had approximately 200 full-time employees. An implicit vision emerged in the discussions between Geely and Volvo that CEVT could become more than a joint technology development center for Geely Auto and Volvo. Geely, despite its global ambitions for Geely Auto, lacked knowledge of successfully conducting its business in the

West and faced a steep learning curve. Commenting on this, one of the respondents stated that "CEVT is the ideal learning context and the future foundation for Geely's business in Europe and globally (Deputy CEO, CEVT, 2015). Furthermore, as CEVT was mandated to develop a vehicle that initially would be sold on the Chinese market, there was a need to understand the particular attributes of the Chinese market and Chinese customers. Accordingly, it was acknowledged at an early stage that "CEVT needs the Western experience, but must not lose the Chinese perspective." (VP, R&D, 2015). It was assessed by CEVT that inclusion of inputs from Chinese resources and Geely Auto's home market was necessary to reach the ambitious goal of setting up functioning organizations in both Sweden and China, develop a platform, and engage in complete vehicle development. One interviewee alluded that "Many of us here at CEVT Gothenburg know Volvo, but we don't know China. Our Chinese colleagues bring their China experience into this. That is pivotal to make this work." (Senior Manager, Vehicle Integration, CEVT, 2015).

CEVT employs top engineers from around the globe, although the majority originates from Sweden and China. The increase in the number of employees since CEVT's opening in September 2013 can be attributed to the relative ease of finding and employing the requisite competence (Ny Teknik, 2015, authors' translation). Many employees have previous experience working with large original equipment manufacturers (OEMs) in Europe, which compensates for Geely Auto's lack of experience. Whereas the current average tenure for a Volvo employee in Sweden is 17 years, the average is only 8 years at Geely Auto, with the majority of engineers being recent graduates with approximately 4 years of work experience (VP HR, Volvo, 2015; Talent Manager, Volvo, 2015). Being located near Volvo also means that CEVT has attracted experienced engineers from Volvo's own R&D department. At its inception, CEVT "borrowed" many engineers from Volvo on two-year contracts. After the completion of their tenure, these engineers returned to Volvo. However, the strong interest in joining CEVT gave rise to concerns that the large number of engineers moving back and forth from Volvo to CEVT might affect the regular business at Volvo (Deputy CEO, CEVT, 2015; VP HR, Volvo, 2015). Subsequently, no Volvo employee was employed at CEVT unless the employee left Volvo altogether and came to work exclusively for CEVT.

For the future, and with further support and funding from the Chinese government, Geely plans to establish an innovation and automotive-supplier park near CEVT Sweden. This strategic intention aims to turn CEVT into a pivotal unit in the automotive industry in the West.

Talking about this issue, a respondent said: “CEVT will act as a benchmark for the Chinese government in its future support of Chinese firms establishing themselves in the West. We [CEVT] are not only developing a product for the Chinese industry. We are also trying to bring a lot of Chinese suppliers together, to raise the Chinese image.” (Deputy CEO, CEVT, 2015).

4.2. Experience, know-how, and organizational growth

Technology development at CEVT focuses on the CMA, a new, small, and flexible modular platform architecture for next-generation Volvo and Geely Auto C-segment vehicles (same size as Audi A3, Ford Focus, and Volvo V40). CEVT's responsibilities cover vital aspects of passenger vehicle development – from the joint architecture to powertrain and driveline components, top hat engineering, and the vehicles' exterior design. Its four areas of expertise are Architectural Development (i.e., creating new modular architectures and key components for C-segment vehicles), Top Hat Development (i.e., creating complete, customer-focused vehicles based on the new architectures), Shared Component Development (i.e., creating new solutions that apply to both Volvo and Geely Auto brands and customer profiles), and Complete Vehicle Design (i.e., creating designed vehicles that expand customer segments) (Senior Manager, Vehicle Integration, CEVT, 2015). Through the CMA platform, Volvo and Geely Auto remain in full control of their next-generation C-segment vehicles and share technology without jeopardizing brand integrity. While Volvo is a premium brand that combines innovation and leadership in safety, Geely Auto is a volume manufacturer targeting multiple segments in China and international markets (Deputy CEO, CEVT, 2015). The modular platform technology will bring a number of benefits that decrease costs and potentially increase profits for both Volvo and Geely Auto. By transferring knowledge between products that are perceived to have the potential to differentiate between cost levels, a platform with multiple attributes can be developed. These attributes include (1) a vertical platform that enables focus on market segments ranging from more to less expensive products, (2) the ability to satisfy different market and brand model demands, (3) incorporation of different customer demands, and (4) reduction of overall development and production costs. This is achieved by developing technological solutions that meet global standards and allow for different levels of vehicle performance based on market and brand. The modular technology also enables easy and quick adaptation to new customer preferences and trends, creates tailor-made solutions for both Geely Auto and Volvo, and meets the requirements of short product cycles and shorter time to market.

In line with custom vehicle platform development and product planning, vehicle top hats (i.e., vehicle body and body interior contents) would be developed separately by Volvo and Geely Auto's own R & D departments. However, after Geely Chairman Li Shufu visited CEVT in December 2013, the growth of the organization surged: “Having made firsthand observations of the achievements reached in a short time regarding competence development and delivery of technical solutions, more responsibility and mandates were given to CEVT.” (Deputy CEO, CEVT, 2015). Shortly after, in mid-2014, Geely Auto ordered 11 complete vehicle models, including conceptual development, design, new technology, top hats, and interiors. Thus, within one year of its establishment, CEVT was not only responsible for developing the joint CMA platform but also had full vehicle development responsibility. By early 2015, responsibilities had evolved to span all aspects of the vehicle value chain, including Product Strategy and Planning, Program Financing, R&D, and Purchasing/sourcing. Commenting on the responsibility given to CEVT, one of the respondents said: “We have become a small OEM, without a factory.” (Deputy CEO, CEVT, 2015). Furthermore, it was only a matter of time before Powertrain, Manufacturing, Marketing, Sales, and After-Market Sales Service would be added to CEVT responsibilities for all Geely Auto CMA business-related models in Europe and the Americas.

This created a need for steep R&D capacity growth and a quick scaling up of human resources and competencies. In 2013, the company was the fastest growing company in terms of the number of employees in the Gothenburg region. By the end of 2014, the number of employees had risen to 260 full-time engineers and about 600 consultants. By autumn 2015, CEVT employed 1100 people, 400 directly by CEVT and another 700 consultants, and had a budget of RMB 2500 million (i.e., a 625 times increase in budget since its inception in 2013). In addition, CEVT had ambitions to employ approximately 1500 people working on the next generation of platforms, engines, and transmissions by end of 2015. According to the CEO of CEVT, he had never experienced such an aggressive product and model plan before: “It has happened more in one year than in the 20 years I spent with Saab. [...] We are no longer just an R&D center.” (Ny Teknik, 2015, authors' translation). One respondent reflected that “This development of CEVT is a significant sign of trust from Geely HQ.” (Director, Corporate communications, Geely, 2014).

4.3. Leverages for innovation capabilities through networks and co-dependence

Drawing on the technological developments used in Volvo's state-of-the-art new vehicle platform, the Scalable Platform Architecture (SPA), CEVT has developed the CMA platform much faster than anticipated. CMA is just like the SPA platform flexible to form the base for several models, bodies, and powertrains. In fact, the CMA is a smaller version of the SPA. In addition, the entire electrical system of CMA is replicated from the SPA. According to one respondent, the ability by CEVT to draw upon Volvo technology “[...] allows us to move a whole and complete electrical architecture into the CMA. And that means that we have many things already established when starting the projects. We try to follow the Volvo architecture as much as possible.” (Senior Manager, Exterior & Body, CEVT, 2015).

This co-development with Volvo aims to modernize both the vehicles and the production processes. Extensive use of common modular components allows Geely to streamline assembly and benefit from economies of scale. This further means that vehicles based on CMA have some commonality with SPA vehicles. For instance, the brand-new Volvo XC40, which is the first Volvo vehicle developed on the CMA platform, has technology commonalities with the new CX90, including Internet connectivity, active and passive safety, transmissions, and powertrains. Volvo also plans to introduce a new S40 and a V40 model based on the CMA platform.

CEVT's vision is that increased technology integration will facilitate strengthening the long-term competitiveness of Geely Auto and Volvo through the development of world-class vehicle architectures, new technologies, and the raising of overall product quality. By merging the technology strengths of Volvo and the cost advantages of Geely Auto at CEVT, Geely hopes to optimize resource utilization, speed up the implementation of a shared platform strategy, increase standardization, and share modularization in product development within its two automotive companies. Commenting on this vision, one interviewee stated that “CEVT works as a bridge between Geely Auto and Volvo to further enhance global competence development synergies between the two companies.” (Director, Corporate communications, Geely, 2014). Desired outcomes are synergies in product quality, brand image, and cost advantages that will provide Geely with strong, sustained, and unique competitive advantages over its competitors in technology, quality, and brand areas (Geely Annual Report, 2013, 2014). Geely's new ambition is that it will engage in the development of new proprietary technology rather than in reverse engineering and technology copying. This will be done by drawing on Volvo's knowledge and expertise in technology and safety. Important learning areas also include development processes, work methodologies, and how to conduct global business in the automotive industry. As a direct result of the strategic initiative to develop a new vehicle platform and a number of

derivative vehicles, a new manufacturing plant for small C-segment vehicles will be established in Taizhou, with a scheduled opening date in 2017 timed with the launch of the first C-segment models. This plant will be owned by Geely Auto but managed and operated by Volvo and will produce both Volvo and Geely Auto vehicles.

In an attempt to increase coordination between CEVT and Geely Auto, an integrated functional entity was established within Geely Research Institute (GRI) in Hangzhou for Geely Auto CMA vehicle projects. GRI is the R&D house of Geely Auto, comprising about 2000 young engineers engaged in R&D. Approximately 400 employees at GRI in China were initially assigned to be directly involved in R&D work for CMA. Referred to as “CEVT China” (Deputy CEO, CEVT, 2015), this new organizational entity was not formally a part of CEVT Sweden but rather set up as an entity within GRI that was led by CEVT. In 2014, however, CEVT China was established through an organizational separation from GRI China, becoming a *de facto* Chinese branch of CEVT. As one respondent said: “Much of Geely Auto’s proprietary technology development will remain at GRI in China, where CEVT will co-operate on the design and development of complete vehicles.” (Deputy CEO, CEVT, 2015).

As a further means of consolidating operations, shared components for future CMA vehicles will be sourced jointly between Volvo and Geely Auto. By combining parts of the sourcing requirements of Volvo and Geely Auto at CEVT (roughly 1 million units), it will be possible to capitalize on volumes, attain a stronger negotiation position vis-à-vis suppliers, make the cost advantage of Geely Auto accessible to Volvo, and subsequently bring cost benefits to the whole Geely Group. Setting up a joint purchasing team between Volvo and Geely Auto, based at CEVT in Gothenburg, is also an important contributing factor to the innovation potential of CEVT as this team is responsible for sourcing all developed material for the coming CMA models, thereby ensuring technological quality for both brands. For example, one respondent said: “Since Chinese engineers are used to working more closely with component suppliers, they can bring valuable experience in terms of cost solutions. One strength of Chinese engineers is that they are better equipped and more experienced in terms of identifying where different cost elements reside in the design of components.” (Director, Purchasing, 2015). Through a new and faster way of developing vehicles, it is possible to combine the elements of premium vehicles with new ideas about cost-efficient solutions, thus providing new opportunities.

In addition, through joint sourcing for Volvo and Geely Auto CMA vehicles, it is also possible to enhance R&D and technology capability, organizational capability, design capability, and supply chain competence in both home markets. To handle the paradox between managing a premium brand (Volvo) and volume brand (Geely Auto) at the same time, the Chinese way of working with suppliers must adhere to and follow Volvo quality methods and standards more closely when sourcing. It was jointly decided that a potential supplier of CMA vehicle-related components had to have R&D activities in Europe while also having a manufacturing factory in China. This makes it possible to meet both quality standards and cost targets. Since Geely had an established strength in China when it came to negotiating with local suppliers, the CEVT steering committee decided that Chinese employees would take the lead in supplier negotiations.

4.4. Organizing for relative absorptive capacity, learning, and knowledge

CEVT is governed by a board of directors consisting of Volvo and Geely representatives. The board is rather small, but significant to the importance that Geely attaches to CEVT. The CEVT chairman of the board is likewise the chairman of Geely. Other group members include the CEO of Volvo, the vice president and chief financial officer (CFO) of Geely, the chief advisor to Geely owner Mr. Li Shufu, and the CEO of CEVT.

Organizationally, CEVT is divided into a number of departments

that are led by a vice president (VP). CEVT’s first line of management has eight departments: R&D, CMA Product Strategy & Vehicle Line Management, Quality, Finance, HR, Purchasing, Geely Design (Gothenburg), and the Business Office. Geely Design is the global design department, with offices located in Shanghai, Barcelona, Los Angeles, and Gothenburg. With the exception of the Shanghai office, the other three offices are legally controlled by CEVT. The Geely Design office in Gothenburg is located merely a five-minute walk from the CEVT Sweden office and employs nearly 300 design engineers. Thus, an integrative connection exists between CEVT Gothenburg, Geely Design, and Geely Global Design (via Geely Design) and Geely Research Institute (via CEVT China).

The R&D department is made up of nine functions: Architecture, Powertrain Integration, Chassis & Safety Electronics, Interior, Exterior & Body, Electrical, Vehicle Integration, Transmission, and Manufacturing Engineering. Altogether, the senior management team consists of 17 VPs with an accumulative 300 years of experience in the automotive industry. This experience constitutes important value for Geely since learning is enhanced by drawing on the vast experience of the automotive industry that exists in Gothenburg. Reflecting upon this, one interviewee stated that “The CEVT organization is built to mimic the Volvo Cars organization because we believe in their way of management and control.” (Director, Exterior & Body, CEVT, 2015). What more, a Senior Manager stated “At CEVT we work according to Volvo’s Global Product Development System, with slight modifications to fit our own organization. There are many employees around here that are well accustomed to working with this development process.” (Senior Manager, CEVT, 2015).

Development work for Geely Auto vehicles is divided almost equally between CEVT Sweden and CEVT China. The first phase, conceptual vehicle development and detailed engineering, is led by CEVT Sweden. The second phase, the launch phase, including prototype development, physical testing, and ensuring quality prior to assembly, is increasingly led by employees at CEVT China since a new testing facility is being built there (Interview, Director, Exterior and Body, CEVT, 2015). Division of responsibilities also depends on the nature of the technology being developed. Detailed engineering work for technology-intensive areas that require extensive dialogue with Vehicle Integration and Design (e.g., lighting) are located in Sweden because these two functions also reside there. Other areas require closer co-operation with suppliers and subsequently detailed engineering work can be carried out by CEVT China. In other technology areas, detailed engineering activities have been transferred to CEVT China midway through the development process, and when required competence has been deemed existing in CEVT China. Hence, detailed engineering development activities are carried out at both CEVT Gothenburg and CEVT China to varying degrees. Nonetheless, the technology lead responsibility always resides with an employee at CEVT Sweden and all technology development results are reported back to CEVT Gothenburg. In order to further develop CEVTs innovative capability, the connection with CEVT China and the resources residing there is actively emphasized. This includes a constant emphasis on the need for a holistic perspective of the development work being carried out at CEVT, and to establish structures, processes, and routines at CEVT China that are similar to those of CEVT Gothenburg. To arrange mutual exchange visits were seen as an important facilitator for this development. As one respondent put it “When we move from the first phase to the launch phase, it is important that the Chinese organization has acquired an insight of what we have done, how we have done it, how we think, and why so. This makes their job easier and hopefully allows them to internalize our knowledge better.” (Senior Manager, Vehicle Integration, CEVT, 2015).

CMA platform development work is governed by a steering committee that is involved in making larger strategic technology decisions. The chairman of the steering committee is a CEVT board member and senior advisor to Zhejiang Geely Holding Group owner Mr. Li Shufu. Other members include executive management team members from

Volvo, Geely Auto, and Geely. Technology development is carried out in an integrated functional organization that includes current CEVT functions (Architecture, Powertrain, Chassis & Electronic Safety, Interior, Exterior & Body, Vehicle Integration, Purchasing, R&D Quality, Manufacturing Engineering, and Design) as well as Prototype and Testing and Homologation Program Management and Vehicle Programs for both Volvo and Geely Auto. Each functional area within a vehicle project is led by a project leader from CEVT Gothenburg and a deputy project leader from CEVT China. Technology development work is carried out in six different cross-functional module teams: Powertrain Integration, Exterior & Body, Interior, Chassis & Safety Electronics, Electrical, and Vehicle Integration. These cross-functional module teams consist of both Swedish and Chinese engineers and play a key role within the CMA organization since they have full release responsibility for delivering the module within defined targets. The module team is led by a module team leader and work proceeds according to standardized work processes. Each module team can, in turn, comprise a number of program module sub-team leaders and part module sub-team leaders. Thus, Exterior & Body can, for instance, be regarded as a program module team consisting of a number of different program module sub-teams, such as front structure, midrange board, forward and backward floor, covering sheet metal, instrument panel, and parts. Each module sub-team is responsible for delivering solutions that comply with the requirements and characteristics of each respective brand. Each module sub-team consists of resources from Manufacturing, Engineering, Quality, Product Design (Styling), Testing/Attribute, and Purchasing. The module team setup also has a duality of roles and representatives. The global module team leader is the person with the most competence. Currently, most global leaders are from Sweden and CEVT Gothenburg, taking global leadership and leading the whole global team. In each sub-module team, however, each CEVT representative is matched by a Geely Auto representative with the same function, role, and responsibility.

Intensive collaboration between Sweden and China is mainly based on staff mobility. CEVT China regularly sends Chinese engineers to CEVT Sweden to learn from their international peers for periods of three months to two years. Currently, around 130 people working at CEVT Sweden are employees of the CEVT China team at GRI in Shanghai, China, and have been transferred to Gothenburg for training and learning. Thus, CEVT provides Geely with access to technology and knowledge by funneling relevant technology competence, local and international talent, intellectual property, and production and platform technology knowledge through CMA. Hence, as one interviewee stated: “An important mission for CEVT is to facilitate learning and knowledge transfer between its employees, especially between the more experienced Western employees and less experienced Chinese employees.” (Deputy CEO, CEVT, 2015). In fact, one of the key performance indicators for Swedish global module leaders is to train their Chinese counterparts and deliver results with the Chinese team (Director, Exterior & Body, CEVT, 2015). Learning is not a one-way process as the Chinese counterparts contribute with important knowledge and learning as well. The Chinese automotive market is currently the most important market in the world, with an estimated customer capacity base of 23 million units. Talking about this issue a respondent said: “Chinese employees at CEVT and Geely Research Institute hold a lot of knowledge and understanding of their home market's requirements with regard to vehicle features in demand.” (Director, Exterior & Body, Geely, 2015). Another respondent meant that “The Chinese employees are also knowledgeable of the value they bring to the close co-operation between Volvo and Geely Auto at CEVT.” (Senior Manager, Vehicle Integration, CEVT, 2015).

5. Discussion

The purpose of this study is to analyze qualitatively how the innovation capabilities and strategic asset-creation of an emerging market

automotive firm is augmented by its FDI location and utilization of a Western target firm's technological portfolio. The innovation capabilities and asset-creation is studied through an in-depth longitudinal case study, which shows the inherent complexity of asset creation initiatives while highlighting the organizational imperatives used by the involved firms. The findings of the study are related to the issue of innovation in the East Asian automotive industry by investigating how automotive industry firms from East Asian emerging markets (in our case China) strive for increased innovation capabilities in order to globalize their product development activities to create innovation for competing on the global and domestic market.

The willingness of Chinese firms to acquire competence and other important assets has accelerated during the last decade (Mathews, 2006; Ström and Nakamura, 2014). Two interconnected events opened up new opportunities in European business around 2008/2009. First, the financial crisis in the U.S. started to severely affect firms in Europe. Second, the Chinese economy showed signs of resilience, which opened up government-supported acquisitions abroad. This is the context in which the Geely acquisition of Volvo should be understood. In this paper, the implications for innovation capabilities among the Chinese acquirers have been put into focus. In other words, what aims did Geely have for its acquisition of Volvo, and what implications in terms of innovation capabilities did this acquisition have for Geely?

Expanding its regional innovation networks to include European firms and production units, Geely has been globalizing and represents a concrete example of the strength in the production and industrial ownership integration over the Eurasian continent. Volvo, then owned by Ford, was put up for sale in 2009 due to bad profitability. After a complex sales process with several potential buyers, Geely ended up with the winning bid. The Geely acquisition of Volvo was also one of the first real tests of a Chinese firm buying a well-known brand name with high-end technical capabilities and distribution channels, spearheading the increased FDI from Asia into Europe (Alvstam et al., 2014). In this acquisition process, the 2013 setup of the new joint research center in Volvo's home base, Gothenburg, was instrumental to the development of Geely's innovation capabilities. The purpose of CEVT is to serve both Volvo and Geely Auto with the development of cars in the small car segment.

In the following sections, the implication of this study for the existing innovation literature is discussed. Subsequently, implications for practice will be presented. Limitations of this study and suggestions for further studies will conclude the chapter.

5.1. Theoretical implications

This study makes three important contributions to the innovation literature in general, and the field of strategic asset-augmentation research in particular.

In the M&A literature, asset-augmentation has been defined as activities carried out to generate new knowledge that augments existing competencies and strengthens competitive advantage within the *acquiring* firm (Buckley et al., 2016). Our *first* core contribution in our study is the results that underline the multifaceted approach Geely used when engaging in *strategic asset-creation*. Thus, the findings of our study provide a single, yet novel, insight into a phenomenon that is yet to be discussed at any length in the existing asset-augmentation literature. We define strategic asset-creation, in the context of M&As in the automotive industry, as the carrying out of new and/or old activities, in the most favorable regional setting, to create technological innovativeness, increase R&D capacity, and create new business value propositions that can benefit *both* the acquired *and* the acquiring firm. Strategic asset-creation is achieved by drawing on existing technological resources, innovation capabilities, and management knowledge from both the acquired and the acquiring firm. In our study, the creation of CEVT and the development of the new inter-firm vehicle platform exemplify the creation of such a strategic asset. More specifically,

CEVT and the vehicle platform constitute a long-term and durable asset that ensures further co-dependence and co-operation between the acquirer and acquiring firm. The shared vehicle platform enables, for instance, Geely Auto to tap into the technological know-how of Volvo and the existing regional competence that exists in the western region of Sweden for the remaining lifespan of the platform. By ensuring that Volvo has a vested interest in the CMA platform and is required to maintain, enhance, and support the continuous development of the CMA platform, Geely in general and Geely Auto, in particular, can draw continuous advantages of regional innovation dynamics to enhance their own innovation capabilities. At the same time, Volvo can gain benefits from Geely's established advantages in China in terms of cost advantages emanating from existing relationships with suppliers, purchasing negotiations, and vehicle assembly, as well as draw advantage of Geely's knowledge of Chinese market demands. Thus, strategic asset-creation by *intertwining existing innovation capabilities of two firms* allows the creation of positive lock-in effects that can aid in keeping firms on well-established industrial and technological trajectories.

The cross-border post-M&A activity, in the context presented here, extends Narula's (2006) and Buckley et al.'s (2016) discussions and is interpreted as investing firms' ambitions to create long-lasting networks and relations between the relative innovative capabilities that exist in the two companies. Our results from the CEVT case can also be seen as adding, for example, to the theses of Caloghirou et al. (2004) and Ferreras-Mendéz et al. (2015) regarding the importance of organizations effectively internalizing innovative capabilities rather than just acquiring the knowledge per se.

Our *second* major contribution is that this study provides a more comprehensive explanation of the importance of spatial dynamics of regional attractiveness in the strategic asset-creation processes of EMNCs. Our study argues that the location decision for CEVT was deliberately made to take advantage of the clustering of industry-related innovation competencies, employee competence, and regional competitiveness found within the automotive cluster in western Sweden, in conjunction with the country-specific advantages derived from the Chinese market. This exemplifies the creation of strategic assets by taking advantage of existing regional dynamics of growth. This is something that has been an important part of the knowledge-driven economy (Lindqvist and Sölvell, 2011; Ström and Wahlqvist, 2010). Our study shows that through leveraging the strength of both companies across regional spheres (i.e., the technological know-how, safety, and brand of Volvo vs. factor cost advantages, China market access, and the sales network of Geely Auto), CEVT manages to enhance global innovation capability synergies between the two companies and builds up competence that aids in ensuring the future success of both automotive brands. Prior studies discussing asset-augmentation have treated the home external environment as a push-factor in EMNCs' quest for asset-augmentation or seen it as a matter of giving rise to country-specific advantages for the EMNCs (see Buckley et al., 2016; Wei, 2010). Subsequently, the existence of firm-specific capabilities at the target firm has been argued to be sufficient for asset-augmentation to occur. However, our findings have important implications for our understanding of the role played by existing firm-specific capabilities at the target firm. While we confirm the sufficiency of existing firm-specific capabilities for asset-augmentation (e.g., in the form of innovation capabilities), we argue that it is a necessary but insufficient condition for the creation of new strategic assets. *Rather, it is the combination of existing firm-specific capabilities together with favorable macroeconomic conditions, at both the target and the acquiring firm, that lay the foundation for the ability to create new strategic assets.* In particular, this finding emphasizes the necessity of favorable spatial dynamics, the influence of host and home country geography and regional innovation dynamics, and the will to optimize resources over space and utilize different location advantages in the strategic asset-creation process. The combination of locational advantages at different locations facilitates the strategic asset-creation process, thereby contributing to raising the

acquirer and target firms' overall innovation capability.

Our *third* contribution responds to calls for further research on how EMNCs harness, absorb, and augment existing assets or create new assets (Anderson et al., 2015; Narula, 2010; Rugman and Li, 2007). The findings in this study give new insights into how EMNCs can organize the creation of new strategic assets to overcome challenges associated with differences in relative absorptive capacity, as Lane and Lubatkin (1998) discuss. This includes how to manage the lack of experience in R&D, foreign M&As, and internal managerial knowledge, as well as the lack of high-skilled labor and high-tech knowledge that characterize many EMNCs (Shenkar, 2009; Gugler and Vanoli, 2015). Our results shown that setting up organizational routines, structures, procedures, and processes in the new strategic asset that are *aligned* with the strategic intents of the acquiring and target firms, but *not determined* by them, contributes to Lane and Lubatkin's (1998) discussion on the needs to circumvent limitations, i.e., limitations emanating from differences in relative absorptive capacity, experience deficits in R&D activities and internal managerial knowledge, and differences in relatedness. The creation of CEVT has allowed for less constrained means to combine existing strengths, develop new knowledge, and exploit existing competence. Such an approach further points to the multifaceted nature of innovation capability creation sought by Geely. Strategic asset-creation transcends the implicit facets of technology development to include other important conditions for dynamics in innovation, such as the emulation of internal organization of R&D activities, organizational structures, work methodologies, procedures, processes, and routines. In addition, as shown in our study, organizational, technological, institutional, and geographic dimensions influence the notion of relatedness differences, and thus also the notion of relative absorptive capacity (see Boschma and Frenken, 2011). The study also connects to the discussion of the development of relational economic geography to make use of networks encapsulating firms and regional actors (Bathelt and Glückler, 2003). To overcome such differences, location-specific learning seems to be a decisive factor. Localized and contextualized learning bear the merits of exposing employees not only to explicit knowledge, but also, perhaps more importantly, to tacit knowledge through forms of relationship building, socialization, and interaction. We argue that tacit technological know-how is needed for both the interpretation and application of augmented or newly created innovation capabilities (i.e., *what* to do and *how* to do it) and that this tacit knowledge requires close access to regional innovation dynamics where much such knowledge is developed. Consequently, tacit knowledge becomes the "glue and integrating mechanism in learning" (Dhanaraj et al., 2004, p. 430) based on the fact of "being there" at specific locations (Gertler, 2003). Table 3 summarizes the main contributions of this study in relation to dynamics in strategic asset-creation in the automotive industry.

5.2. Implications for practice

This research generates a number of takeaways for practitioners involved in the management of a strategic asset-creation process in the automotive industry. *First*, decision-makers and managers need to take measures to ensure that a new strategic asset retains a high degree of mandated autonomy. Such autonomy should be achieved by ensuring that the intention with the new strategic asset is aligned with, but not determined by, the strategies of the acquiring firm or the target firm. Autonomy is likely to give rise to strategic flexibility, allowing managers to make more independent decisions regarding operational activities, human capital strategies, technology strategies, product portfolio configurations, and division of labor. *Second*, managers must be cautious about dynamic adjustment costs that are most likely to occur as a result of the need to progressively and increasingly adjust productive resources during the establishment and growth stage. Such costs and disruptions may emanate from the need to hire new employees, transfer existing managerial resources from the acquired and acquiring firm, or be a result of limits to the scope, range, and

Table 3
Main contributions - Dynamics in strategic asset-creation in the automotive industry.

Factor	Derived from	Sources
Firm-level capabilities	Acquiring firm Target firm	- Existing technological resources - R&D capacity - Existing human capital - Management knowledge
Macro-level conditions	Spatial dynamics (<i>home country</i>) Spatial dynamics (<i>host country</i>)	- Country-specific advantages - Automotive industry-related innovation competencies - Regional competitiveness - Automotive cluster competence - Regional dynamics of growth - Proximity and availability of needed competence
Synergies	Leverage of existing strengths and advantages	- Increased co-dependence, co-operation, and co-ordination - Positive lock-in effects - Long-lasting networks and relations - Technology integration/technology sharing - Business acumen
Dissemination	Organizational configuration	- Emulation of internal organization of R & D activities, organizational structures, way-of-working, procedures, processes, routines - Exploiting existing competence - Knowledge transfer - Developing new knowledge - Localized and contextualized learning
Innovation upgrading	Strategic Asset-Creation	The carrying out of (new) activities, in the most favorable regional setting, to create technological innovativeness, increase R&D capacity, and create new business value propositions that are beneficial to both the acquired and the acquiring firm. Strategic asset-creation is achieved by drawing on existing technological resources, innovation capabilities, and management knowledge from both the acquired and the acquiring firm.

complexity of tasks that can be performed by the new entity's existing stock of human capital. In particular, there is a need to exercise extra caution with disruptions that can not only cascade within the new entity but also spill over to the acquiring firm or the target firm. Therefore, managers should put mechanisms in place that ensure that the new entity's capacity to build, integrate, reconfigure, and deploy organizational resources through its existing stock of human capital grows concurrently with the increasing demand for coordination of various resources, competencies, and capabilities. *Third*, the relative strengths and weaknesses emanating from country-specific advantages and firm-specific advantages can shift during the evolution of the new strategic asset. Hence, managers should recognize that during the development of the strategic asset, the acquiring firm, the target firm, or as shown in this case the new strategic asset, can and should be adaptive to take on the role as both teacher and student (Lane and Lubatkin, 1998).

From a regional policy perspective, important lessons can be drawn from this case. Strong companies do not grow independently; they need, for example, strong supplier networks, skilled labor, and distribution channels. It is clear from this case that Chinese investors are seeking firms that can improve their own long-run securement of innovative capability and facilitate knowledge transfer. These capabilities can be obtained at different levels. The CEVT case shows that the technical development over time has become highly connected to the regional knowledge center and cluster advantages of agglomeration (Asheim et al., 2011; Ström and Wahlqvist, 2010). Technical spillovers, global distribution chains, and a strong brand name are all examples of how this can be achieved. This study also points toward the complexity of creating regional competitive advantage related to a variety of industrial and service sub-sectors, which is a result of industrial development over a sustained period of time. As a result, which Geely's decision to establish CEVT in Gothenburg suggests, the establishment of science parks and other programs combining education and research are important in private firm location strategy. Thus, active public sector involvement through local, regional, and even national support programs can play an important role vital to obtaining regional economic resilience and sustained competitive regional advantage.

Policy development needs to facilitate a firm's growth and secure a base of highly skilled labor in both manufacturing and services. The

complexity of value-added within the automotive industry calls for a combination of policies to take advantage of related industries within a region. These structural preconditions can be facilitated by establishing regional innovation networks actively engaged regional development agencies at the intersection of public and private sectors and individual firms and local political structures. Well-functioning networks are difficult to imitate and thus become a competitive advantage for a specific region.

5.3. Limitations and suggestions for further research

The study concerns one single case of a Chinese automotive firm acquiring a Swedish auto manufacturer. Such a research design implies some important limitations. First and most obviously, the generalizability of our findings might be limited as not all insights from this particular case can be applied to all international M&As in the automotive industry. Second, other influential factors might have been missed by only studying one case and, therefore, alternative research designs might make the findings clearer. These limitations lead us to suggest ways to develop the research in this field.

Despite the limitations of this study, its findings still point to a number of interesting venues for further development of research on how organizational dynamics can be understood as an important factor to advance the innovative capabilities of both an acquiring firm and a target firm in cross-border M&As. A concrete suggestion to further the research into the mechanisms behind strategic *asset-creation*, which we propose differs from strategic *asset-augmentation*. Acquiring organizational capabilities, which are intangible in nature (as compared to tangible, physical, innovations leading to, e.g., improved production technologies and products), through a cross-border M&A might yield a strategy that EMNCs can use to rapidly enhance their own innovative capabilities. Thus, studies on other, similar, cases of cross-border M&A where the acquiring firm is an EMNC would be important contributions to the research field.

6. Conclusion

Different from the general pattern of Chinese outward M&A, where investors are often state-owned enterprises (e.g., Ren et al., 2012),

private investors have dominated the Chinese cross-border M&A initiatives in Northern Europe during the post-2000 period (Zephyr, 2015). Still, the Geely acquisition of Volvo aligns with the Chinese government's designated list of strategic industries of national interest. This case demonstrates the strength of asset-seeking motives aiming for technology enhancement, acquisition of know-how and patents, and access to well-established brand names of acquired foreign firms, which all characterize the CEVT establishment. Our findings are not only in line with what is observed earlier by, e.g., Hong and Sun (2006) and Ström and Nakamura (2014), but also show how asset-creation opportunities and strengthened innovation capabilities emerge through cross-border M&As. In this respect, our contribution to the innovation literature shows how the integration of firm-specific advantages or capabilities, and regional advantage can facilitate innovation leverages that build stronger competitive advantages. Thus, the findings from this study also broaden the scope of understanding OFDI beyond production motives by pointing to wider industrial dynamics such as organizational characteristics vital for innovation capability enhancement.

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